Home Work-1

1) Sum of Two Matrices.

```
#include <stdio.h>

int main() {
    int a[10][10], b[10][10], sum[10][10];
    int r, c, i, j;

    printf("Enter rows and columns: ");
    scanf("%d%d", &r, &c);

    printf("Enter first matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&a[i][j]);

    printf("Enter second matrix:\n");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&b[i][j]);

    printf("Sum Matrix:\n");
    for(i=0;i<r;i++) {
        for(j=0;j<c;j++) {
            sum[i][j]=a[i][j]+b[i][j];
            printf("%d ",sum[i][j]);
        }
        printf("\n");
    }
    return 0;
}
```

Output:

```
Enter rows and columns:
2 2
Enter first matrix:
1 2
3 4
Enter second matrix:
5 6
7 8
Sum Matrix:
6 8
10 12
```

2) Write a program C to read n number of values in an array & display them in reverse order.

```
#include <stdio.h>

int main() {
```

```
    int a[100], n, i;

    printf("Enter number: ");
    scanf("%d",&n);

    printf("Enter elements:\n");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Reverse Order:\n");
    for(i=n-1;i>=0;i--)
        printf("%d ",a[i]);

    return 0;
}
```

Output:

```
Enter number:
5
Enter elements:
10 20 30 40 50
Reverse Order:
50 40 30 20 10
```

3) Count the total number of duplicate elements in an array.

```
#include <stdio.h>

int main() {
    int a[100], n, i, j, count=0;

    printf("Enter number: ");
    scanf("%d",&n);
    printf("Enter elements:");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    for(i=0;i<n;i++) {
        for(j=i+1;j<n;j++) {
            if(a[i]==a[j]) {
                count++;
                break;
            }
        }
    }

    printf("Duplicate elements = %d",count);

    return 0;
}
```

Output:

```
Enter number:
10
Enter elements:
10 20 30 20 50 70 10 80 50 95
Duplicate elements = 3
```

4) Print unique elements in an array.

```c
#include <stdio.h>

int main() {
    int a[100], n, i, j, count;

    printf("Enter number: ");
    scanf("%d",&n);
    printf("Enter elements: ");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Unique Elements:\n");

    for(i=0;i<n;i++) {
        count=0;
        for(j=0;j<n;j++) {
            if(a[i]==a[j])
                count++;
        }
        if(count==1)
            printf("%d ",a[i]);
    }

    return 0;
}
```

Output:

```
Enter number:
5
Enter elements:
10 78 35 28 78
Unique Elements:
10 35 28
```

5) Insert in Sorted Array.

```c
#include <stdio.h>

int main() {
    int a[100], n, i, item;

    printf("Enter number: ");
    scanf("%d",&n);
    printf("Enter elements: ");
    for(i=0;i<n;i++)
```

```
        scanf("%d",&a[i]);

    printf("Enter insert element: ");
    scanf("%d",&item);

    i=n-1;

    while(i>=0 && a[i]>item) {
        a[i+1]=a[i];
        i--;
    }

    a[i+1]=item;
    n++;

    printf("Sorted array:\n");

    for(i=0;i<n;i++)
        printf("%d ",a[i]);

    return 0;
}
```

```
Output:

Enter number:
6
Enter elements:
108 127 144 162 178 189
Enter insert element:
112
Sorted array:
108 112 127 144 162 178 189
```

6) Insert in unsorted array.

```
#include <stdio.h>

int main() {
    int a[100], n, pos, item, i;

    printf("Enter number: ");
    scanf("%d",&n);
    printf("Enter elements: ");
    for(i=0;i<n;i++)
        scanf("%d",&a[i]);

    printf("Enter position: ");
    scanf("%d",&pos);
    printf("Enter insert element: ");
    scanf("%d",&item);

    for(i=n;i>=pos;i--)
        a[i]=a[i-1];
```

```
    a[pos-1]=item;
    n++;
    printf("Unsorted array:\n");
    for(i=0;i<n;i++)
        printf("%d ",a[i]);

    return 0;
}
```

```
Output:

Enter number:
4
Enter elements:
21 34 56 87
Enter position:
2
Enter insert element:
88
Unsorted array:
21 88 34 56 87
```

7) Matrix multiplication.

```
#include <stdio.h>

int main() {
    int a[10][10], b[10][10], c[10][10];
    int r1,c1,r2,c2,i,j,k;

    printf("Enter rows and cols of first matrix: ");
    scanf("%d%d",&r1,&c1);

    printf("Enter rows and cols of second matrix: ");
    scanf("%d%d",&r2,&c2);

    if(c1!=r2) {
        printf("Multiplication not possible");
        return 0;
    }

    printf("Enter first matrix:\n");
    for(i=0;i<r1;i++)
        for(j=0;j<c1;j++)
            scanf("%d",&a[i][j]);

    printf("Enter second matrix:\n");
    for(i=0;i<r2;i++)
        for(j=0;j<c2;j++)
            scanf("%d",&b[i][j]);

    for(i=0;i<r1;i++) {
        for(j=0;j<c2;j++) {
```

```
            c[i][j]=0;
            for(k=0;k<c1;k++)
                c[i][j]+=a[i][k]*b[k][j];
        }
    }

    printf("Matrix multiplication:\n");

    for(i=0;i<r1;i++) {
        for(j=0;j<c2;j++)
            printf("%d ",c[i][j]);
        printf("\n");
    }

    return 0;
}
```

```
Output:

Enter rows and cols of first matrix:
 2 2
Enter rows and cols of second matrix:
 2 2
Enter first matrix:
1 2
3 4
Enter second matrix:
1 2
3 4
Matrix multiplication:
7 10
15 22
```

8) Matrix transpose.

```
#include <stdio.h>

int main() {
    int a[10][10], t[10][10];
    int r,c,i,j;
    printf("Enter row and column: ");
    scanf("%d%d",&r,&c);
    printf("Enter matrix: ");
    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            scanf("%d",&a[i][j]);

    for(i=0;i<r;i++)
        for(j=0;j<c;j++)
            t[j][i]=a[i][j];

    printf("Matrix Transpose:\n");

    for(i=0;i<c;i++) {
```

```
        for(j=0;j<r;j++)
            printf("%d ",t[i][j]);
        printf("\n");
    }

    return 0;
}
```

Output:

```
Enter row and column: 3 3
Enter matrix:
11 22 33
44 55 66
77 88 99
Matrix Transpose:
11 44 77
22 55 88
33 66 99
```

9) Right diagonal sum.

```
#include <stdio.h>

int main() {
    int a[10][10], n, i, j, sum=0;
    printf("Enter number: ");
    scanf("%d",&n);
    printf("Enter matrix: ");
    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&a[i][j]);

    for(i=0;i<n;i++)
        sum+=a[i][i];

    printf("Right diagonal=%d",sum);

    return 0;
}
```

Output:

```
Enter number:
3
Enter matrix:
101 102 103
104 105 106
107 108 109
Right diagonal=315
```

10) Check for identity matrix.

```
#include <stdio.h>
```

```
int main() {
    int a[10][10], n, i, j, flag=1;
    printf("Enter number: ");
    scanf("%d",&n);
    printf("Enter matrix: ");
    for(i=0;i<n;i++)
        for(j=0;j<n;j++)
            scanf("%d",&a[i][j]);

    for(i=0;i<n;i++) {
        for(j=0;j<n;j++) {
            if(i==j && a[i][j]!=1)
                flag=0;
            if(i!=j && a[i][j]!=0)
                flag=0;
        }
    }

    if(flag)
        printf("Identity Matrix");
    else
        printf("Not Identity Matrix");

    return 0;
}
```

Output:

```
Enter number: 3
Enter matrix:
1 0 0
0 1 0
0 0 1
Identity Matrix
```

11) To find number of vowels,consonants,digits and white spaces.

```
#include <stdio.h>

int main()
{
    char str[100];
    int i,vowels = 0, consonants = 0;
    int digits = 0, spaces = 0;

    printf("Enter a string: ");
    fgets(str,sizeof(str),stdin);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == 'a' || str[i] == 'e' || str[i] == 'i' ||
           str[i] == 'o' || str[i] == 'u' ||
           str[i] == 'A' || str[i] == 'E' || str[i] == 'I' ||
```

```
            str[i] == 'O' || str[i] == 'U')
        {
            vowels++;
        }
        else if((str[i] >= 'a' && str[i] <= 'z') ||
                (str[i] >= 'A' && str[i] <= 'Z'))
        {
            consonants++;
        }
        else if(str[i] >= '0' && str[i] <= '9')
        {
            digits++;
        }
        else if(str[i] == ' ')
        {
            spaces++;
        }
    }

    printf("Vowels = %d\n", vowels);
    printf("Consonants = %d\n", consonants);
    printf("Digits = %d\n", digits);
    printf("White spaces = %d\n", spaces);

    return 0;
}
```

Output:

```
Enter a string: Machine 123
Vowels = 3
Consonants = 4
Digits = 3
White spaces = 1
```

12) Frequency of a character in a string.

```
#include <stdio.h>

int main()
{
    char str[100], ch;
    int i, count = 0;

    printf("Enter a string: ");
    fgets(str,sizeof(str),stdin);

    printf("Enter character: ");
    scanf("%c", &ch);

    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ch)
            count++;
```

```c
    }

    printf("Frequency = %d", count);

    return 0;
}
```

Output:

```
Enter a string: banana
Enter character: a
Frequency = 3
```

13) Count words in a string.

```c
#include <stdio.h>

int main()
{
    char str[100];
    int i, count = 1;

    printf("Enter a string: ");
    fgets(str,sizeof(str),stdin);



    for(i = 0; str[i] != '\0'; i++)
    {
        if(str[i] == ' ')
            count++;
    }

    printf("Number of words = %d", count);

    return 0;
}
```

Output:

```
Enter a string: apple
Number of words = 1
```

14) Sort string array.

```c
#include <stdio.h>
#include <string.h>

int main()
{
    char str[10][50], temp[50];
    int n, i, j;

    printf("Enter number of strings: ");
```

```
    scanf("%d", &n);

    printf("Enter strings:\n");

    for(i = 0; i < n; i++)
        scanf("%s", str[i]);

    for(i = 0; i < n - 1; i++)
    {
        for(j = i + 1; j < n; j++)
        {
            if(strcmp(str[i], str[j]) > 0)
            {
                strcpy(temp, str[i]);
                strcpy(str[i], str[j]);
                strcpy(str[j], temp);
            }
        }
    }

    printf("Sorted strings:\n");

    for(i = 0; i < n; i++)
        printf("%s\n", str[i]);

    return 0;
}
```

Output:

```
Enter number of strings: 4
Enter strings:
eye
nose
ear
hair
Sorted strings:
ear
eye
hair
nose
```

15) Toggle case of a sentence.

```
#include <stdio.h>
#include <ctype.h>

int main() {
    char str[100];

    printf("Enter a sentence: ");
    fgets(str, sizeof(str), stdin);

    for (int i = 0; str[i] != '\0'; i++) {
```

```
        if (islower(str[i]))
            str[i] = toupper(str[i]);
        else if (isupper(str[i]))
            str[i] = tolower(str[i]);
    }

    printf("Toggle sentence: %s", str);

    return 0;
}
```

```
Output:

Enter a sentence: education
Toggle sentence: EDUCATION
```